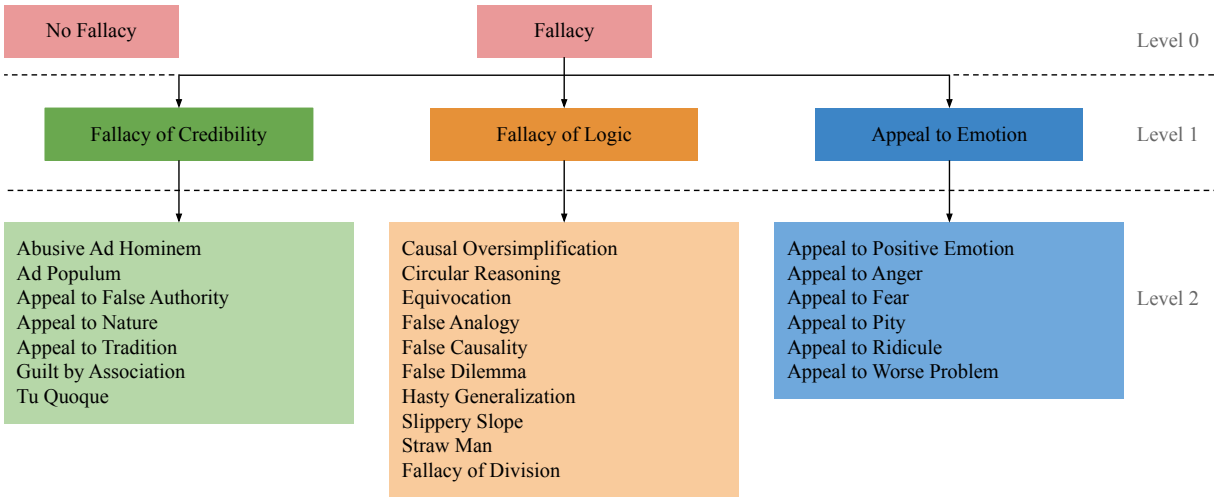

Figure 2: Tree structure of our taxonomy. Detailed definitions of the fallacies are in Appendix A.

many emotions), fallacies that appeared in only a single work (e.g., *confusion fallacy* appears only in Martino et al. (2019)), and we merged fallacies that were too similar in their definitions (like *begging the question* and *circular reasoning*). Some fallacies were not taken into account because they were not actually fallacies in our definition. These are, e.g., rhetorical techniques such as *flag waving* or *repetition*. Our list can obviously be extended in the future with new fallacies. Details on how our collection unifies existing works are in Appendix B.

We grouped our fallacies into broader categories to create a taxonomy on top of our collection. We chose the categories that Aristotle originally proposed (Wisse, 1989), because it has been shown to be applicable across various forms of communication – from political speeches to advertisements (Wisse, 1989). This yields the following taxonomy:

1. **Level 0** is a binary classification, categorizing text as either fallacious or non-fallacious.
2. **Level 1** groups fallacies into Aristotle's categories: 'Pathos' (appeals to emotion), 'Ethos' (fallacies of credibility), and 'Logos' (fallacies of logic, relevance, or evidence).
3. **Level 2** contains fine-grained fallacies within the broad categories of Level 1. For instance, under *fallacy of credibility*, we have specific fallacies such as *appeal to tradition*, *ad populum*, and *guilt by association*.

Our taxonomy is shown in Figure 2. For each fallacy, we provide **both a formal and an informal definition** in Appendix A (inspired by Bennett (2012)). For instance, the *appeal to ridicule* is informally defined as "an argument that portrays the opponent's position as absurd or ridiculous with the intention of discrediting it.". Formally, it is defined as "E_1 claims P. E_2 makes P look ridiculous, by misrepresenting P (P'). Therefore, $\neg P$.", where E_i are entities (e.g., people, organizations, etc.), and P is a proposition.

4 Tackling Subjectivity in Annotations

4.1 Subjectivity in Fallacy Annotation

Annotating fallacies is an inherently subjective endeavor. To see this, consider Example (c) in Figure 1. The argument goes that the candidate has to win again because he won last time. This can be seen as a *false causality* fallacy: a cause-effect relationship is incorrectly inferred between two events that have nothing to do with each other. However, it can also be seen as a *causal oversimplification* fallacy. This is because we can contend that having won the last election gives the candidate an edge over other candidates in terms of visibility, and thus makes it more likely that he wins this year's election as well. The argument is thus fallacious mainly because it fails to acknowledge other factors that play a role in re-election.

This simple example already shows subjectivity in fallacy annotations, where several annotations can be defended. It would be counter-productive if the annotators converged on, say, *causal oversimplification*, so that every approach of fallacy annotation is penalized for proposing an (equally plausible) *false causality*. There are other cases of legitimately differing opinions: One annotator may see implicit assertions that another annotator does not see. In "*Are you for America? Vote*

for me!" one reader may see the implicit "*or you must be against America*" (which makes this a *false dilemma*), while another annotator may see no such implicature. Annotators may also have different thresholds for fear (*appeal to fear*) or insults (*ad hominem*). Finally, different annotators have different background knowledge: A sentence such as "*Use disinfectants or you will get Covid-19!*" may be read as a plausible warning by one annotator but as an *appeal to fear* fallacy by an annotator who knows that Covid-19 does not spread via contaminated surfaces. We will now present a disjunctive annotation scheme that accounts for this inherent subjectivity.

4.2 Disjunctive Annotation Scheme

Before presenting our annotation scheme, we need to establish some common ground:

Definition 4.1: Text

A text is a sequence of sentences $st_1, \ldots, st_n$.

A *span* on a text is a contiguous sequence of sentences. The set S of all spans of a text $st_1, \ldots, st_n$ is thus $S = \{st_i \ldots st_j \mid 0 < i \leq j \leq n\}$.

Definition 4.2: Span

The span of a fallacy in a text is the smallest contiguous sequence of sentences that comprises the conclusion and the premises of the fallacy. If the span comprises a pronoun that refers to a premise or to the conclusion, that premise or conclusion is not included in the span.

We work on the level of sentences, because previous work has shown that agreement on the token level is even harder to achieve (Jin et al., 2022). We allow the use of pronouns to decrease the size of the spans: When a sentence refers to another sentence by a pronoun, that other sentence does not have to be part of the span. For instance, in Example (d) of Figure 1, the premise of the fallacy is in the title of the post, and the conclusion is at the end of the text. Thus, a span that covers the entire fallacy would have to cover the entire post from title to end. However, the pronoun "This" refers to the title, and thus we can omit the title from the span. Nevertheless, a span can comprise several sentences.

A span can be annotated with a label (such as a fallacy type) by an annotator (or a group of them) or by a system. We now propose the key element of our disjunctive annotation scheme, in which subjectivity is not projected away, but explicitly embraced by allowing for several equally valid labels for the same span.

Definition 4.3: Gold Standard

Let $\mathcal{F}$ be the set of fallacy types and $\perp$ be a special label that means "no fallacy".
Given a text and its set of spans S, a gold standard G is a set of pairs of a span $s \in S$ and a set of labels from $\mathcal{F} \cup \{\perp\}$:

$$G \subseteq S \times (\mathcal{P}(\mathcal{F} \cup \{\perp\}) \setminus \{\emptyset, \{\perp\}\})$$

Here, $\mathcal{P}(\cdot)$ denotes the powerset.

The gold standard associates a given span with one or more fallacy labels. If more than one label is present, this means that any label is acceptable as an annotation. The gold standard can also associate a span with $\perp$, which means that the annotation of this span is optional. However, in this case, the gold standard has to associate the span also with at least one other label, as we are not interested in annotating non-fallacious sentences. The gold standard can also contain the same span twice, which means that the span has to be annotated with two labels. The alternative labels for a span can be generated through various methods during the annotation process, e.g., one annotator giving alternatives, a group of annotators proposing different labels due to lack of consensus, or multiple independent annotators combining their labels (see Example 4.1).

We define a prediction as the annotation of a text by a system or a user:

Definition 4.4: Prediction

Given a set of fallacy types $\mathcal{F}$, a text, and its set of spans S, a prediction P is a set of pairs of a span $s \in S$ and a label $l \in \mathcal{F}$:

$$P \subseteq S \times \mathcal{F}$$

The following example gives substance to these definitions:

Example 4.1

Let "$a\ b\ c\ d$" be a text where a, b, c, and d are sentences.
Suppose $S = \{a\ b, d\}$ (i.e., the sentences a and b are one fallacious span, and d is a span of one fallacious sentence), $a\ b$ has labels $\{l_1, l_2\}$, and d has label $\{l_3\}$. In that case, $G = \big\{(a\ b, \{l_1, l_2\}), (d, \{l_3\})\big\}$

An example of prediction P could be $P = \{(a, l_1), (a, l_2), (b, l_3), (c, l_4), (d, l_1)\}$

4.3 Evaluation Metrics.

To compare two annotated spans, we adapt the precision and recall of Martino et al. (2019) to alternatives. Given two spans, p with its label l_p, and g with its set of labels l_g, respectively, and a normalizing constant h, these metrics compute a comparison score as follows:

$$C(p, l_p, g, l_g, h) = \frac{|p \cap g|}{h} \times \delta(l_p, l_g)$$

δ is a similarity function. We use $\delta(x, y) = [x \in y]$, where $[\cdot]$ is the Iverson bracket.

Let G be the gold standard, and let P be the prediction of a user or a system. The precision for P is computed by comparing each span in P against all spans in G, and taking the score of the best-matching one:

$$\textit{Precision}(P, G) = \frac{\sum_{(p,l_p)\in P} \max_{(g,l_g)\in G} C(p, l_p, g, l_g, |p|)}{|P|}$$

If there are no annotations in P (i.e., $|P| = 0$), we set precision to 1. This choice is inspired by the intuition that a loss in precision should result only from false predictions. If there are no such false predictions, then precision should not be harmed.

To calculate the recall, we exclude all the spans from the gold standard that contain a $\perp$. The rationale for this choice is that when a span has also been marked as "no fallacy", its annotation is considered optional. Therefore, we do not want to penalize models that do not provide an annotation for such a span. We define the set G^-, which is G restricted to the spans that do not map to $\perp$, i.e., $G^- = \{(s, L) \in G \mid \perp \notin L\}$. The recall is then computed as:

$$\textit{Recall}(P, G) = \frac{\sum_{(g,l_g)\in G^-} \max_{(p,l_p)\in P} C(p, l_p, g, l_g, |g|)}{|G^-|}$$

If $|G^-| = 0$, we set the recall to 1. The intuition is that a model should be penalized in recall only for the annotations it misses from the gold standard. If there are no such missed annotations, recall should not suffer. Appendix L shows how our metrics handle various edge cases.

The F1-score is computed as usual as the harmonic mean of precision and recall. It is easy to see that our definitions of precision and recall fall back to the standard definitions if G does not have alternatives and all spans comprise only a single sentence. In that case, the score C is 1 if the spans are identical and identically labeled. If there are no alternatives, our measures are also identical to the ones in Martino et al. (2019), with one difference: we use the $\max$ instead of a sum in the definitions to select the best matching span. In this way, two neighboring spans with the same label do not achieve full precision or recall if the gold standard requires one contiguous span with that label. Using $\max$ instead of a sum also avoids the case where Martino et al. (2019)'s metric yields precision or recall scores exceeding one. This occurs when the gold standard contains overlapping spans with identical labels, affecting precision, or when the prediction includes such overlaps, affecting recall. However, their metric is equivalent to ours as long as (1) there are no alternatives in the gold standard, (2) neither the gold standard nor the prediction contains overlapping spans with the same label and (3) each span from the gold standard overlaps with at most one span with the same label from the predictions and vice versa. Examples, proofs, and further details are in Appendix I.

5 MAFALDA Dataset

5.1 Source Datasets

We used four publicly available fallacy datasets to construct our benchmark:

We imported all 8,576 texts from Sahai et al. (2021). These are online discussions from Reddit, as in Example (d) of Figure 1. We reconstructed the texts by concatenating the post of interest, the previous post (if available), and the title. The title was considered as a citation and was thus not annotated. This dataset contains sentences that were labeled as negative examples.

We imported all 336 texts from Martino et al. (2019), which are from news outlets. We imported all 583 texts from Jin et al. (2022), which are either toy examples gathered from online quizzes (as in